

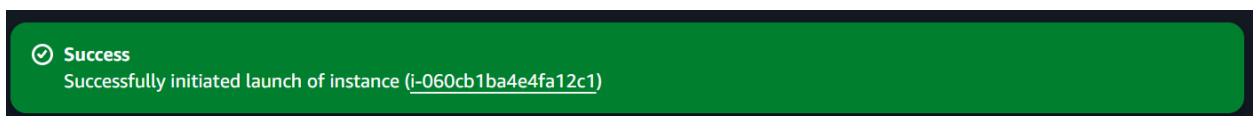
Task -15

15-7-26

Topic : Deploying a GitHub Project on AWS EC2 Using the Public IP Address

Step 1: Launch an EC2 Instance

1. Log in to the AWS Management Console.
2. Open EC2 Dashboard.
3. Click Launch Instance.
4. Enter the instance name
5. Select Ubuntu Server 24.04 LTS
6. Choose c7i-flex.large
7. Create or select an existing key pair.
8. Under Network Settings:
 - Allow SSH (Port 22).
 - Allow HTTP (Port 80).
 - Add Custom TCP (Port 8080) if your application uses port 8080.
9. Click Launch Instance.



Step 2: Connect to the EC2 Instance

Open your terminal and run:

```
ssh -i "rrr.pem" ubuntu@ec2-3-101-59-11.us-west-1.compute.amazonaws.com
```

Step 3: Update the Server:

Command to update: `sudo apt update`

Step 4: Install Git

Command : `sudo apt install git -y`

Verify: `git --version`

Step 5: Install Java

Install OpenJDK 17:

Command : `sudo apt install openjdk-17-jdk -y`

Verify: `java -version`

Step 6: Install Maven (if your project uses Maven)

Command for install: `sudo apt install maven -y`

Verify the install run the command : `mvn -version`

Step 7: Clone the GitHub Repository

Copy the repository URL from GitHub.

Run the command : `git clone https://github.com/username/project.git`

Check that it was cloned: `ls`

Move into the project directory: `cd project name`

Step 8: Build the Project

If it is a Maven project:

Command for see : `mvn package`

After the build completes, check the generated JAR:

Command : `ls target`

```

ubuntu@ip-172-31-8-37:~$ ls
spring-petclinic
ubuntu@ip-172-31-8-37:~$ cd spring-petclinic/
ubuntu@ip-172-31-8-37:~/spring-petclinic$ ls
LICENSE.txt  build.gradle  gradle  gradlew.bat  mvnw  pom.xml  src
README.md   docker-compose.yml  gradlew  k8s  mvnw.cmd  settings.gradle  target
ubuntu@ip-172-31-8-37:~/spring-petclinic$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.springframework.samples:spring-petclinic >-----
[INFO] Building petclinic 4.0.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- enforcer:3.6.3:enforce (enforce-java) @ spring-petclinic ---
[INFO] Rule 0: org.apache.maven.enforcer.rules.version.RequireJavaVersion passed
[INFO]

```

Step 9: Run the Application

Run the command : **java -jar spring-petclinic-4.0.0-SNAPSHOT.jar**

```

ubuntu@ip-172-31-8-37:~/spring-petclinic/target$ ls
checkstyle-cachefile  classes  maven-archiver  spring-petclinic-4.0.0-SNAPSHOT.jar.original
checkstyle-checker.xml  generated-sources  maven-status  surefire-reports
checkstyle-header.txt  generated-test-sources  site  test-classes
checkstyle-result.xml  jacoco.exec  spring-petclinic-4.0.0-SNAPSHOT.jar  test-ids
ubuntu@ip-172-31-8-37:~/spring-petclinic/target$ java -jar spring-petclinic-4.0.0-SNAPSHOT.jar

```



```

:: Built with Spring Boot :: 4.1.0

```

Step 10: Configure the EC2 Security Group

1. Open EC2 Dashboard.
2. Select your instance.
3. Go to the Security tab.
4. Click the Security Group.
5. Choose Edit Inbound Rules.
6. Add a rule:

Step 11: Get the Public IP

Go to the EC2 instance.

Copy the **Public IPv4 Address**.

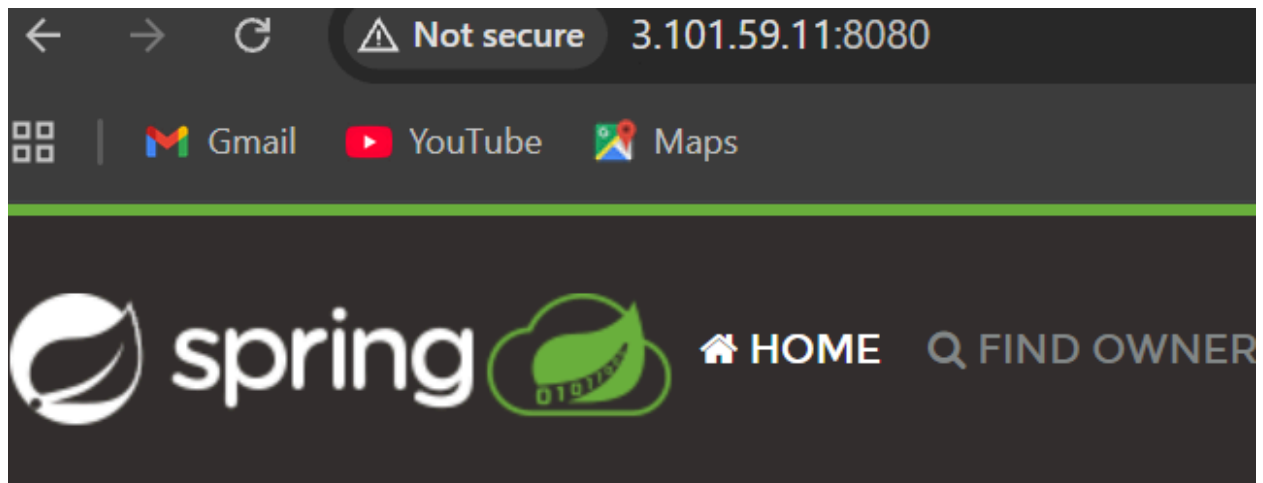
Step 12: Access the Application

If your application runs on port 8080:

3.101.59.11:8080

Step 13: Verify the Application

The application will open in your web browser.



Welcome



